



Year 11 Extension 1 online Assessment Task 3

Questions 41-44 (20 marks)

Please start each question on a new page.

41.

A pot of soup is to be refrigerated overnight. It needs to be cooled from its temperature of  $100^{\circ}\text{C}$  before it can be placed in the fridge.

The pot of soup is cooled in a sink full of cold water kept at a temperature of  $5^{\circ}\text{C}$ . The temperature of the pot of soup,  $T^{\circ}\text{C}$ , after  $t$  minutes in the sink satisfies the equation

$$\frac{dT}{dt} = -k(T - 5).$$

(i) Show that  $T = 5 + 95e^{-kt}$  satisfies both this equation and the initial condition. **1**

(ii) After 10 minutes in the sink, the temperature of the pot of soup reduces from  $100^{\circ}\text{C}$  to  $60^{\circ}\text{C}$ . It must be cooled to a temperature of  $20^{\circ}\text{C}$  before it can be placed in the fridge.

How long (to the nearest minute) must the pot of soup remain in the sink full of cold water? **3**

42. a.

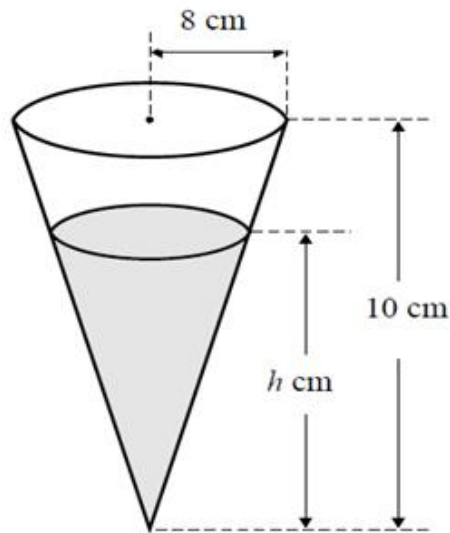
Find the exact term independent of  $x$  in the expansion of

$$(1 + 2x)^2 \left(2x + \frac{1}{x}\right)^{12} \quad 2$$

b. Use the binomial expansion of  $(1 + x)^n$ , to prove that

$$\binom{n}{1} + 4\binom{n}{2} + 12\binom{n}{3} + \cdots + r\binom{n}{r}2^{r-1} + \cdots + n\binom{n}{n}2^{n-1} = n \cdot 3^{n-1} \quad 2$$

43.



The figure above shows an inverted conical cup with base radius 8 cm and height 10 cm.

Some water is poured into the cup at a constant rate of  $\frac{2\pi}{5}$  cm<sup>3</sup> per minute.

Let the depth of the water be  $h$  cm at time  $t$  minutes.

4

Find the rate of change in the area of the water surface when  $h = 4$

44. Given that  $m = a(\cos x)^3 + 3a \cos x (\sin x)^2$

and  $n = a(\sin x)^3 + 3a \sin x (\cos x)^2$

Prove that  $(m + n)^{\frac{2}{3}} + (m - n)^{\frac{2}{3}} = 2a^{\frac{2}{3}}$

4

### Q41

i,  $T = 5 + 95e^{-kt}$  (\*)

$\therefore T - 5 = 95e^{-kt}$  (1)

$$\frac{dT}{dt} = -k \times 95e^{-kt}$$

$$\frac{dT}{dt} = -k(T-5), \text{ from (1)}$$

so (\*) satisfies the equation

$t=0 \rightarrow T = 5 + 95e^{-k \times 0}$

$$= 5 + 95$$

$$\therefore T = 100$$

so (\*) satisfies the initial condition.

(1)  
need BOTH

ii, when  $t=10$ ,  $T=60$

$$\therefore 60 = 5 + 95e^{-k \times 10}$$

$$55 = 95e^{-10k}$$

$$\frac{11}{19} = e^{-10k}$$

$$-10k = \ln\left(\frac{11}{19}\right)$$

$$k = \frac{1}{10} \ln\left(\frac{19}{11}\right)$$

(1)

we want  $t$  when  $T=20$

$$20 = 5 + 95e^{-kt}$$

$$\frac{3}{19} = e^{-kt}$$

$$-kt = \ln\left(\frac{3}{19}\right)$$

$$t = \frac{1}{k} \ln\left(\frac{19}{3}\right)$$

$$= \frac{10 \ln\left(\frac{19}{3}\right)}{\ln\left(\frac{19}{11}\right)}$$

$$t = 33.77 \rightarrow t = 34 \text{ mins.}$$

(1)

∴ must remain for a total of 34 mins

OR

must remain for an additional 24 mins.

① final

## Question 42

Tuesday, 12 October 2021 12:39 pm

Find the exact term independent of  $x$  in the expansion of

$$(1+2x)^2 \left(2x + \frac{1}{x}\right)^{12}$$

$$\begin{aligned} \text{a) } & (1+2x)^2 \left(2x + \frac{1}{x}\right)^{12} \\ &= (1+4x+4x^2) \left(2x + \frac{1}{x}\right)^{12} \\ &= (1+4x+4x^2) \sum_{r=0}^{12} {}^{12}C_r (2x)^{12-r} (x^{-1})^r \\ &= (1+4x+4x^2) \sum_{r=0}^{12} {}^{12}C_r 2^{12-r} x^{12-2r} \end{aligned}$$

For term independent of  $x$ :

$$\text{either } 12-2r = 0 \quad \text{or} \quad 12-2r = -2$$

$$[12-2r \neq 1 \text{ since } r \in \mathbb{Z}^+]$$

$$\therefore r = 6 \quad \text{or} \quad r = 7$$

$\therefore$  term independent of  $x$

$$= 1 \times {}^{12}C_6 \cdot 2^6 + 4 \times {}^{12}C_7 \cdot 2^5$$

$$= 160512$$

1 mark for  $r=6$  and  $r=7$

or for  $r=6$  and term

1 mark for total value of term.

Use the binomial expansion of  $(1+x)^n$ , to prove that

$$\binom{n}{1} + 4\binom{n}{2} + 12\binom{n}{3} + \dots + r\binom{n}{r}2^{r-1} + \dots + n\binom{n}{n}2^{n-1} = n \cdot 3^{n-1}$$

$$\text{b) } (1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \dots + {}^nC_r x^r + \dots + {}^nC_n x^n$$

Differentiating both sides:

$$n(1+x)^{n-1} = {}^nC_1 + 2{}^nC_2 x + 3{}^nC_3 x^2 + \dots + r{}^nC_r x^{r-1} + \dots + n{}^nC_n x^{n-1}$$

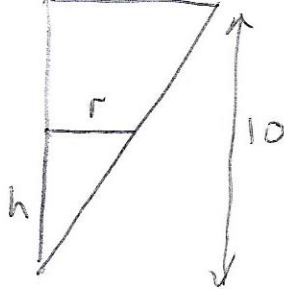
Let  $x=2$ :

$$n \cdot 3^{n-1} = {}^nC_1 + 2{}^nC_2 \cdot 2 + 3{}^nC_3 \cdot 2^2 + \dots + r{}^nC_r \cdot 2^{r-1} + \dots + n{}^nC_n \cdot 2^{n-1}$$

$$\therefore {}^nC_1 + 4{}^nC_2 + 12{}^nC_3 + \dots + r{}^nC_r \cdot 2^{r-1} + \dots + n{}^nC_n \cdot 2^{n-1} = n \cdot 3^{n-1}$$

1 mark for differentiating both sides with explanation

1 mark for substituting  $x=2$  with explanation and showing final statement.



$$\frac{r}{h} = \frac{8}{10} \therefore r = \frac{4h}{5} \quad \checkmark$$

$$\frac{dv}{dt} = \frac{dv}{dh} \times \frac{dh}{dt} \therefore \frac{dh}{dt} = \frac{dv/dt}{dv/dh}$$

$$V = \frac{1}{3} \pi r^2 h = \frac{16}{75} \pi h^3 \quad \checkmark$$

$$\frac{dv}{dh} = \frac{16}{25} \pi h^2 \quad \text{and} \quad \frac{dv}{dt} = \frac{2\pi}{5} \text{ (given)}$$

$$\frac{dh}{dt} = \frac{2\pi/5}{\frac{16}{25} \pi h^2} = \frac{5}{8h^2} \quad \checkmark$$

$$A = \pi r^2 = \frac{16}{25} \pi h^2$$

$$\frac{dA}{dt} = \frac{dA}{dh} \times \frac{dh}{dt}$$

$$= \frac{32}{25} \pi h \times \frac{5}{8h^2} = \frac{4\pi}{5h} \quad \checkmark$$

$$\text{For } h=4, \quad \frac{dA}{dt} = \frac{\pi}{5}$$

$$V = \frac{1}{3} Ah = \frac{1}{3} A \frac{5A}{4\pi}^{1/2} = \frac{5}{12\sqrt{\pi}} A^{3/2}$$

$$\frac{dv}{dA} = \frac{5}{8\sqrt{\pi}} A^{1/2} = \frac{5}{8\sqrt{\pi}} \times \frac{4\sqrt{\pi}h}{5} = \frac{h}{2}$$

$$\frac{dA}{dt} = \frac{dA}{dV} \times \frac{dV}{dt} = \frac{2}{h} \times \frac{2\pi}{5} = \frac{4\pi}{5h}$$

$$h^2 = \sqrt{\frac{25}{16\pi} A} \\ = \frac{5}{4} \sqrt{\frac{A}{\pi}}$$

Q44.

Given that  $m = a(\cos x)^3 + 3a \cos x (\sin x)^2$  and  $n = a(\sin x)^3 + 3a \sin x (\cos x)^2$ . Prove that  $(m + n)^{\frac{2}{3}} + (m - n)^{\frac{2}{3}} = 2a^{\frac{2}{3}}$

$$\begin{aligned}m + n &= a \cos^3 x + 3a \cos^2 x \sin x + 3a \cos x \sin^2 x + a \sin^3 x \\&= a(\cos^3 x + 3 \cos^2 x \sin x + 3 \cos x \sin^2 x + \sin^3 x) \\&= a(\cos x + \sin x)^3\end{aligned}$$

1 mark for simplifying  $m + n$

$$\begin{aligned}(m + n)^{\frac{1}{3}} &= a^{\frac{1}{3}}(\cos x + \sin x) \\(m + n)^{\frac{2}{3}} &= a^{\frac{2}{3}}(\cos x + \sin x)^2 \\&= a^{\frac{2}{3}}(\cos^2 x + \sin^2 x + 2 \cos x \sin x) \\&= a^{\frac{2}{3}} + 2a^{\frac{2}{3}} \cos x \sin x\end{aligned}$$

1 mark for simplifying  $(m + n)^{\frac{2}{3}}$

$$\begin{aligned}m - n &= a \cos^3 x - 3a \cos^2 x \sin x + 3a \cos x \sin^2 x - a \sin^3 x \\&= a(\cos^3 x - 3 \cos^2 x \sin x + 3 \cos x \sin^2 x - \sin^3 x) \\&= a(\cos x - \sin x)^3\end{aligned}$$

$$\begin{aligned}(m - n)^{\frac{1}{3}} &= a^{\frac{1}{3}}(\cos x - \sin x) \\(m - n)^{\frac{2}{3}} &= a^{\frac{2}{3}}(\cos x - \sin x)^2 \\&= a^{\frac{2}{3}}(\cos^2 x + \sin^2 x - 2 \cos x \sin x) \\&= a^{\frac{2}{3}} - 2a^{\frac{2}{3}} \cos x \sin x\end{aligned}$$

1 mark for simplifying  $(m - n)^{\frac{2}{3}}$

$$\begin{aligned}\therefore (m + n)^{\frac{2}{3}} + (m - n)^{\frac{2}{3}} &= \left(a^{\frac{2}{3}} + 2a^{\frac{2}{3}} \cos x \sin x\right) + \left(a^{\frac{2}{3}} - 2a^{\frac{2}{3}} \cos x \sin x\right) \\&= 2a^{\frac{2}{3}}\end{aligned}$$

1 mark for finishing the proof